# Unit – IV: Patents

A patent is a government granted right for a fixed time period to exclude others from making, selling, using, and importing an invention, product, process or design, or improvements on such items. These exclusive, monopoly rights are powerful, and in return the inventor is required to describe the invention in writing. The end result is simply a written description, accompanied by diagrams and drawings, that explains the invention. The public benefits because anyone can read the details of the invention and improve upon it. Importantly, the patent not only allows the public to gain an understanding of the invention, but also defines its limits. Once the patent term expires (generally 20 years from the application filing date), the technology covered by the patent becomes a part of the public domain and is essentially free to use by the public.

## Classification of Patents

Generally, there are three main types of patents: utility patents, design patents and plant patents.

1) **Utility Patent –** When most people think of patents, they are referring mainly to utility patents. A utility patent is a patent that covers inventions, whether it's an innovative software process, a new product that is distinct from prior art, or an improvement to a car engine. A utility patent can be granted for any new, useful, and non-obvious process or product.
2) **Design Patent –** A design patent covers a new and original ornamental design of a product. In other words, a design patent protects the look of a product. Examples of products protected by design patents include jewellery and watches, electronic devices, computer icons, and beverage containers. A design patent consists of numerous drawings that show a product from various angles and contains very little written description, if any.
3) **Plant Patents -** Plant patents were first created by the Patent Act of 1930, which had been proposed by Luther Burbank to protect new species of asexually reproduced plants, mostly flowers. These are different than the utility patents granted to bioengineered plants used in agriculture. The United States was the first country in the world to grant plant patents, and even today many countries continue to deny protection for plants. Indeed, even some signatories to the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS) administered by the World Trade Organization (WTO) reserve the right to deny patents for plants. To be patentable, plants must be cultivated rather than found in the wild, and plant patents are granted only to protect a new, distinct, and non-obvious variety of asexually reproduced plant—i.e., those grown not with seeds but by grafting, budding, or cutting. A plant need not be useful to qualify for a patent, but it must be distinctive in its colour, habit, soil, flavour, productivity, form, or other aspects.

## Patent Search

Patent search is a search of the patent database to determine if there are any patent application similar or identical to an invention that is to be patented. Patent search can be done to improve

the chances of obtaining a patent registration or to find information about new inventions that have patent protection.

Before filing a patent application, a patent search can help with different objectives like:

- ❖ Determining the probability of having a patent granted to a proposed invention.
- ❖ Determining the claims to be filed in the patent application.
- ❖ Determining the freedom to operate.
- ❖ Determining whether a granted patent can be invalidated.
- ❖ Knowing more about similar inventions and status of similar patent filings.

The Seven Steps in a Preliminary Search of U.S. Patents and Published Patent Applications

1. Brainstorm terms to describe your invention based on its purpose, composition and use.

2. Use these terms to find initial relevant Cooperative Patent Classifications using the USPTO website's Classification Text Search Tool (https://www.uspto.gov/web/patents/classification/) Enter the keyword or keywords you wish to search in the Search Tool box. For example, if you were trying to find CPC Classifications for patents related to umbrellas, you would enter "umbrella". The default search system is CPC, Cooperative Patent Classification, so the button "All CPC" is selected. Click on the "Search" button. Scan the resulting classification's Class Schemes (class schedules) to determine the most relevant classification to your invention. If you get zero results in your Classification Text search, consider substituting the word(s) you are using to describe your invention with synonyms, such as the alternative terms you came up with in Step 1.

3. Verify the relevancy of CPC classification you found by reviewing the CPC Classification Definition linked to it (if there is one).

4. Retrieve U.S. patent documents with the CPC classification you selected in the PatFT (Patents Full-Text and Image) database (http://patft.uspto.gov). Review and narrow down the most relevant patent publications by initially focusing on the front-page information of abstract and representative drawings.

5. Using this selected set of most relevant patent publications, review each one in-depth for similarity to your own invention, paying close attention to the additional drawing's pages, the specification and especially the claims. References cited by the applicant and/or patent examiner may lead you to additional relevant patents.

6. Retrieve U.S. published patent applications with the CPC classification you selected in Step 3 in the AppFT (Applications Full-Text and Image) database (http://appft.uspto.gov). Use the same search approach used in Step 4 of first narrowing down your results to the most relevant patent applications by studying the abstract and representative drawings of each on its front page. Then examine the selected published patent applications closely, paying close attention to the additional drawings pages, the specifications and especially the claims.

7. Broaden your search to find additional U.S. patent publications using keyword searching in PatFT or AppFT databases, classification searching of non-U.S. patents on the European

Patent Office's Worldwide Espacenet patent database (http://worldwide.espacenet.com) and searching non-patent literature disclosures of inventions using the free electronic and print resources of your nearest Patent and Trademark Resource Center (http://www.uspto.gov/ptrc).

**Patent search in India**

There is no cost for doing a patent search in India. A patent search can be done through the Patent database of India available at: http://ipindiaservices.gov.in/publicsearch. Depending on the status of a patent application, a patent search can be done under two publication types: published or granted. The user can choose the desired publication type by clicking on the checkbox. The user can view many categories like

- Application Date
- Title
- Abstract
- Complete Specification
- Application Number
- Patent Number
- Applicant Name
- Inventor Name
- Inventor Country
- Inventor Address
- Filing office
- PCT Application Number
- PCT Publication number

The entire category has a drop down box from which the user has access to change the category. There is a search box next to every category where the user can enter the keyword of the patent that he wants to view. By entering a query in more than one box, the applicant can run very precise patent searches. Once the required keywords are entered in the respective boxes, there is a captcha code the user has to clear.

Patent Information

Once the code is entered, there are a number of relevant patent results for the patent search query that the user has entered. On selecting an application, the document opens with the Application Number, Title, Application Date and Status. The user can get further details about the patent by clicking on the Application Number, Title, Application Date and Status.

Through patent search, the applicant can find the following information about the patent:

- ✓ Invention title
- ✓ Publication Number
- ✓ Publication Date
- ✓ Publication Type
- ✓ Application Number
- ✓ Application Filing Date

- ✓ Priority Number
- ✓ Priority Country
- ✓ Priority Date
- ✓ Field of Invention

Classification

Inventor Name, Address, Country, Nationality, Applicant Name, Address, Country, Nationality.

When each column is selected, the patent application's details about that section can be known by the user. When a user selects application number details like Invention Title, Publication number, date, type, etc. There are separate columns that give the user the inventor's and the applicant's Name, Address, Country, and Nationality.

## Patent Ownership:

Patents are items of personal property and thus may be owned, sold, licensed, or devised by will. Until September 16, 2012, applications for patents must be filed by the actual inventor of the article, process, design, or plant. After that date, under the America Invents Act (AIA), applications may be filed by the assignee of the invention (or one to whom the inventor has an obligation to assign), making it easier for the true owner of an invention, often an employer, to file the application. If there is more than one inventor, the application must be signed by all inventors (or assignees, after September 16, 2012).

### Sole Inventors

An invention may be the product of one person or more than one person. If one person conceives of the invention, he or she is the sole inventor. A directive given by an employer to an employee to solve a problem or invent an article does not make the employer an inventor. It is finding a solution to a problem rather than articulating it that determines whether one is an inventor. Similarly, giving suggestions, making minor contributions, or helping to build a model or embodiment of the invention do not make a person an inventor.

### Joint Inventors

When more than one person contributes to an invention, they are joint inventors. Persons may be joint inventors even though they do not physically work together or at the same time, do not make the same type or amount of contribution to the invention, or do not make a contribution to the subject matter of every claim of a patent. 35 U.S.C. § 116. Each, however, must have made some contribution of inventive thought to the resulting product, and there must be some amount of collaboration or connection between them. Thus, individuals who are completely ignorant of what each other has done cannot be considered joint inventors.

## Transfer and Assignment of patent rights:

Because patents have the attributes of personal property, they may be transferred or assigned, just as may other items of personal property, and may be bequeathed by will. Patents or applications for patents may be assigned to another; a written instrument is required. The written instrument should identify the patent by application or issue number, date, and title of the invention. An assignment is a transfer of a party's entire ownership interest or a percentage of that party's ownership interest. A few states have prescribed certain formalities to be observed in connection with the sale of patent rights.

In the absence of agreement to the contrary, each of the joint owners of a patent may make, use, license, offer to sell, or sell the patented invention without the consent of and without accounting to the other owners. Thus, similar to copyright law, each owner has the right to assign or license the patent to a third party without agreement of the other co-owners. Unlike copyright law, however, which requires accounting of profits, patent law does not require a joint owner to account to the others for monies received from such sale. Thus, it is critical that joint inventors be subject to a written agreement detailing their rights to use, make, sell, and license the invention to others.

## Licensing Of Patent Rights

A license differs from an assignment in that it is not an outright grant or transfer of ownership. The licensing of a patent transfers a bundle of rights, which is less than the entire ownership interest. A license is merely a permission to use. The permission or license may be limited in its scope, duration, terms, or territory. A patent license is, in effect, a contractual agreement that the patent owner will not sue the licensee for patent infringement if the licensee makes, uses, offers for sale, sells, or imports the claimed invention, as long as the licensee fulfills its obligations and operates according to the terms of the license agreement. Licenses may be exclusive (meaning that only one party has the ability to exploit the invention) or may be nonexclusive (meaning that more than one party may be given rights in the invention or patent). Similarly, licenses may be for the term of the patent (20 years from the date of filing) or may be for a limited time period.

Licenses may be granted to one party to make the invention and to another party to sell the invention. They may be restricted to specific geographical areas, so that one party has rights for one region of the city to another.

Licenses are not usually recorded with the USPTO because they are viewed as private contractual relationships between parties that do not affect the ultimate ownership of the patent.

The patent owner may charge a one-time lump-sum payment for the license or may receive royalties or periodic payments during the term of the license. Typically, royalty payments are based upon sales of the invention.

## Patent Infringement

## Direct Infringement

A patent issued by the USPTO does not grant a party any right to make, use, offer to sell, sell, or import an invention but rather *excludes* others from engaging in such acts. Under the Patent Act, any person who, without authority, makes, uses, offers to sell, or sells any patented invention within the United States or imports into the United States any patented invention during the term of its patent, infringes the patent.

A patent is effective and enforceable only after it is issued. Thus, making, using, selling, or importing devices prior to the time of a patent's issuance do not constitute acts of infringement.

Liability for patent infringement may arise from one's acts in making, using, or selling the patented invention, referred to as direct infringement, or from encouraging or inducing another to infringe, often referred to as indirect infringement. One critical distinction between the two is that liability can be imposed for direct infringement without regard to the infringer's intent; however, liability for indirect infringement requires that the defendant know that infringement will occur because of his or her conduct. Thus, one can be liable for direct infringement even if one does not know of the existence of the patent infringed.

## Indirect Infringement

A person may be liable for patent infringement even if he or she never makes, uses, or sells the patented invention; liability may rest upon acts of encouraging others to infringe. Such acts are often referred to as indirect infringement. Indirect infringement is covered by two separate subdivisions of the Patent Act:

• whoever actively induces infringement of a patent shall be liable as an infringer.

• whoever offers to sell or sells a nonstable component of a patented invention, knowing it to be especially made or adapted for use in infringement of a patent, shall be liable for contributory patent infringement.

## Remedies For Infringement

A plaintiff who is successful in a suit for patent infringement may obtain injunctive relief, monetary damages, interest, costs of litigation, and, possibly, triple damages and attorneys' fees.

### *Injunctive relief.*

In many instances, a patentee is as interested in ensuring that infringing activities cease as in recovering damages. A court may grant a preliminary injunction pending the final judgment in the case upon a showing that the patentee is reasonably likely to prevail in the action, that irreparable harm would result to the patentee unless an injunction were granted and that monetary damages would be inadequate to compensate for that injury, that the balance of hardships tips in favour of the patentee, and that the public interest would not be disserved by

an injunction. For years, once courts determined that patent infringement had occurred, injunctions were ordered nearly automatically.

### *Compensatory damages.*

The Patent Act provides that upon finding infringement, a court shall award a claimant damages adequate to compensate for infringement, but in no event less than a reasonable royalty for the use made of the invention by the infringer, together with an award of interest and costs.

If a patentee can prove actual damages, such as lost profits, those damages will be awarded. Lost profits will be awarded if the plaintiff can prove that but for the infringement, it would have made the sales that were made by the infringer. If such damages are highly speculative or cannot be proven because the infringer has only made rather than sold the invention, courts will then use the reasonable royalty rate as the measure of damages.

### *Punitive damages and advice of counsel.*

Courts are authorized to increase compensatory damages up to three times the amount assessed. An award of such **punitive damages** (those meant to punish the defendant rather than compensate the plaintiff) is made only upon a showing of willful infringement or bad faith by the defendant. Proof of willful infringement that would permit enhanced damages requires at least a showing of objective recklessness. Accordingly, a patentee must show by clear and convincing evidence that the defendant acted to infringe despite an objectively high likelihood that its actions constituted infringement. The infringer's state of mind is irrelevant. The patentee must then establish that the facts that would satisfy the standard of recklessness were known or should have been known to the alleged infringer.

### *Attorneys' fees.*

A court may award reasonable attorneys' fees to the prevailing party "in exceptional cases." If the patentee is the prevailing party, attorneys' fees might be awarded when the infringer's conduct is wilful or in bad faith. If the accused infringer is the prevailing party, he or she may be able to recover attorneys' fees if the patentee instituted or continued the litigation in bad faith or was guilty of inequitable conduct, either in procuring the patent or in the course of litigation. Attorneys' fees awarded in some cases have reached $1 million.

## New developments in Patent law:

The Patent Act has proven remarkably flexible in accommodating changes and developments in technology. Thus, advancements in technology generally have not necessitated significant changes in the statutes governing patent protection. This situation is somewhat different from that encountered in trademark law with the conflict over domain names creating much turmoil, and in copyright law with the ramifications of publication on the Internet creating similar turmoil.

**Business Method and Software Patents**

Many of the cutting-edge issues in patent law relate to patents for computer software. For several years, the conventional wisdom was that unless a computer program had significant commercial value and application, patent protection was often counterproductive or ineffective in that the USPTO often took more than two years to issue a patent, roughly the same time it took for the software program to become obsolete. Thus, protection of the program under copyright law was viewed as the most effective means of protection for software.

In 1996, guidelines for computer related inventions (often called "software patents") were adopted by the USPTO to aid examiners in determining the patentability of software. Under the guidelines (available at the USPTO website at http://www.uspto.gov), the following are patentable: a machine that is directed by a computer program, a computer readable memory, and a computer-implemented process.

Algorithms that manipulate only abstract ideas remain unpatentable. In brief, patent applications for computer-related inventions must be carefully drafted by attorneys knowledgeable in the art, the developing case law, and the new USPTO guidelines.

**Defensive patenting**

When patents are issued without adequate review of prior art, too many patents are issued, causing technology owners to stumble over each other's intellectual property and producing an avalanche of litigation.

In addition to the drain on resources and time caused by high-stakes patent litigation, numerous companies expend additional sums to patent inventions of questionable use solely for defensive purposes. Defensive patenting, the practice of obtaining patents not because of a desire to exploit the invention but solely to thwart efforts by others who might claim rights in similar methods and processes, has become increasingly common. Some experts refer to this as the patent practitioners' equivalent of mutually assured destruction.

**Patent Trolls**

Patent owners have long used their patents to produce revenue streams by licensing the right to use patented inventions or technology to others. The licensee might pay a flat fee to use the patented technology or might pay a royalty based on the amount of sales of the patented invention or process.

In a twist on patent licensing, numerous companies now secure patents for inventions they have no intention of using or commercializing, solely for the purpose of initiating litigation against claimed infringers. Known as **patent trolls** or non-practicing entities, these companies acquire patents (often from bankrupt companies) and then begin looking or "trolling" for potential infringers. When the patent troll is unable to secure a quick settlement or licensing agreement, it initiates patent litigation, often threatening to obtain an injunction and shut down the other company's business.

**Biotechnology Patents**

Medicine, science, agriculture, and pharmacology present other cutting-edge issues in patent law. Research into genes may hold the key to curing disease throughout the world. Agricultural research may hold the key to providing sufficient food for the world's ever-increasing population. The USPTO continues to see increases in patents in these fields.

In one interesting case, the USPTO issued a patent on Basmati rice lines and grains. The government of India requested re-examination of the patent and submitted 50,000 pages of information to the USPTO arguing that the patent should not have been issued because it was not novel. India was concerned that the patent would damage exports from its own farmers to the United States, a type of "biopiracy" that would carry off the genetic material and biological resources of India. Most of the claims in the patent were declared invalid.

## International patent law

In regard to international patent law, because patents granted in the United States have no effect outside U.S. territorial borders, inventors desiring patent protection in foreign countries must comply with the laws of the countries in which they desire patent protection. Alternatively, however, they may rely on three treaties to which the United States adheres.

Under the Paris Convention, a U.S. inventor who files a patent application in any of the more than 170 Paris member nations has 12 months to file applications in any of the other member nations and yet claim the priority date of the first filing.

Under the Patent Cooperation Treaty (PCT), an inventor may file one "international" application, thereby receiving the benefit of one centralized filing, searching, and examination process and receiving protection in any PCT member nation, as long as the application is ultimately prosecuted in those nations. Prosecution in the other countries may be delayed for up to 30 months, affording the inventor significant time to gather funds, consider the commercial application of the invention, and evaluate market conditions. Applicants may also rely on the European Patent Convention to obtain patent protection in any or all of the 38 contracting nations.

Finally, under the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS), member nations must afford patent protection to citizens of member nations on the same basis they do for their own citizens. Foreign investors may apply for patents in the United States as long as they comply with the provisions of the Patent Act.